Product requirements: AI voice agent for loan servicing

PRD for a proof of concept · Version 0.1 · 5 November 2026 · Language: English (others later)

1 Summary

A voice AI agent for a bank that **calls customers** about upcoming and missed loan payments, and **answers customers who call in** with first-level support. It verifies who it is speaking to, looks up the customer's loans through a secured gateway, explains the bank's policies from a knowledge base, collects structured information (reason for non-payment, promise-to-pay date, callback preference), and hands the call to a human whenever the situation needs one.

The POC runs entirely on one Mac against a **mock bank**: synthetic customers, loans, transactions and policy documents. Its purpose is to prove the conversation design, the tool and data boundary, and the guardrails, not scale or telephony carrier integration.

2 Problem and opportunity

- Early-stage delinquency (1–30 days past due) is mostly forgetfulness, timing or temporary hardship. A timely, polite call resolves most of these, but human call capacity is expensive and limited, so many customers are never called before they become seriously delinquent.
- Inbound lines are dominated by repetitive questions such as "what do I owe?", "when is it due?", "what happens if I'm late?" and "how do I pay?", which a well-grounded agent can answer at any hour.
- Every call should leave structured data behind (reason codes, promises, follow-ups) that is often missing from manual call notes.

3 Goals and non-goals

Goals (POC)

- Hold a natural, interruptible English voice conversation over a SIP phone call.
- Run two outbound campaigns (pre-due reminder and missed payment) and inbound support.
- Answer only from bank data and policy documents, never from model memory.
- Capture structured call outcomes into the mock bank.
- Enforce identity verification, disclosures and contact rules in code.
- Run fully locally with open-source components.

Non-goals (POC)

- Taking payments, or changing balances, terms or contact details.
- Making credit decisions, granting restructuring or waiving fees (the agent can only refer these).
- Real carrier numbers, high call concurrency and multi-language support.
- Integration with a real core banking system or CRM.
- Formal regulatory sign-off.

4 Users

USER	NEEDS
Borrower (customer)	Clear information on what is owed and when, what happens if they pay late and what their options are, and the ability to reach a human. To be treated respectfully and not called excessively.

USER	NEEDS
Collections / support officer	To receive escalated calls with context already captured, and to see promises and callbacks that need follow-up.
Operations / compliance lead	To configure campaigns and contact rules, review transcripts and outcomes, and confirm that disclosures and verification happened on every call.

5 Core use cases

ID	USE CASE	BEHAVIOUR
U1	Outbound: pre-due reminder	Call customers with an instalment due within 3 days. Verify identity, state the amount and due date, offer payment channels, confirm intent to pay and record the outcome.
U2	Outbound: missed payment (1–30 days past due)	Verify, give the required disclosure, state the overdue amount, ask *why* the payment was missed (and record a reason code), explain the consequences from policy (late fee, credit-bureau reporting, next stages), and capture a promise-to-pay date and amount. Refer hardship cases to a human.
U3	Inbound: account questions	Caller asks for balance, next due date, last payment, payoff estimate or loan status. The agent verifies, then answers from the gateway.
U4	Inbound: policy and product questions	"What happens if I pay late?", "Can I pay early?", "How does restructuring work?" Answered from the knowledge base, citing the policy, without requiring verification for general questions.
U5	Follow-up and callback	Schedule a callback at the customer's preferred time. The agent calls back with context, for example about a promise-to-pay that was missed.
U6	Escalation to a human	Triggered by a dispute, hardship, complaint, mention of a lawyer or bankruptcy, a request for a human, two failed verification attempts, or repeated misunderstanding. The agent does a warm transfer with a spoken and written summary, or creates a callback ticket if no human is available.

6 Functional requirements

Priority: **P0** is needed for the POC demo; **P1** is included if time allows.

ID	REQUIREMENT	PRI.
F1	**Voice conversation.** Streaming STT, LLM and TTS with barge-in, end-of-turn detection, and graceful handling of silence (re-prompt, then politely end the call).	P0
F2	**Inbound calls.** Answer calls to the support extension, greet the caller, identify the intent and route it to the right flow.	P0
F3	**Outbound calls.** Place calls from a campaign queue or on demand from the console. Detect no-answer, busy and voicemail, and retry according to rules.	P0
F4	**Identity verification.** Confirm the right party (full name, then date of birth plus the last 4 digits of the loan number) before revealing any account data. Lock after 2 failures. Third parties get only a neutral callback message.	P0
F5	**Customer data via gateway.** Read-only tools for profile, loans, schedule, recent transactions and past interactions, scoped to the verified customer only.	P0
F6	**Policy answers (RAG).** Retrieve from the policy knowledge base and answer with the source document cited in the log. If confidence is low, say so and offer a follow-up instead of guessing.	P0
F7	**Information capture.** Write a structured outcome for every call: contact result, verification result, reason code, promise-to-pay (date and amount), callback request, escalation flag, summary and sentiment.	P0

ID	REQUIREMENT	PRI.
F8	**Compliance guardrails.** AI and recording disclosure at the start of the call; collections disclosure on U2 calls; calling-window, frequency cap, do-not-call and consent checks before dialling; an oral dispute or a "stop calling me" request is honoured immediately.	P0
F9	**Human hand-off.** Warm transfer to a human extension with a summary, or create a callback ticket.	P0
F10	**Operations console.** Trigger campaigns and single calls; view live and past calls, transcripts, tool calls, outcomes and the follow-up queue.	P0
F11	**Audit and tracing.** Record every turn, tool call, retrieved document and latency figure. Redact PII in traces. Each call records the prompt, policy and model versions it used.	P0
F12	**Simulated-caller test suite.** Text-mode scenarios (cooperative, hardship, dispute, wrong person, angry, off-topic, prompt-injection) scored automatically.	P1
F13	**Post-call summary** by SMS or email (mocked as an outbox table).	P1
F14	**Swappable model providers**: local LLM by default, with a hosted LLM switchable by configuration for quality and latency comparison.	P1

7 Conversation design: missed-payment call (U2)

1. **Open:** "Hello, this is the virtual assistant from [Bank]. This call is recorded. May I speak with [first name]?"
2. **Right-party check:** if a third party answers, ask for a good time to call back and end the call with no details given. If the customer is confirmed, verify their date of birth and the last 4 digits of the loan number.
3. **Disclosure:** the configured collections disclosure text, which is versioned.
4. **Purpose:** "Your payment of [amount] due on [date] hasn't been received yet." If the customer says they already paid, check recent transactions and record a payment-in-transit outcome.
5. **Discover:** ask the reason, in an open-ended way, and map it to a reason code (forgot, insufficient funds, income disruption, dispute, other).
6. **Inform:** explain the consequences and timeline from the policy knowledge base: late fee, reporting at 30 days past due, and the next stages.
7. **Resolve:** capture a promise-to-pay (date within the policy window, and amount), or refer hardship or dispute cases to a human.
8. **Confirm and close:** read back the commitment, list the payment channels, and offer a callback or human contact.

8 Data captured per call

`call_id`, `direction`, `campaign`, `customer_id`, `loan_id`, `contact_result` (right_party | third_party | no_answer | voicemail | wrong_number), `verified`, `disclosures_given[]`, `intent[]`, `reason_code`, `promise_to_pay{date, amount}`, `callback{datetime}`, `escalated{reason, target}`, `dispute_flag`, `cease_contact_flag`, `sentiment`, `summary`, `kb_sources[]`, `versions{prompt, policy, model}`, `latency_p50/p90`, `recording_uri`.

9 Non-functional requirements

Latency	Median time from end of speech to first agent audio of 2.0 s or less on the Mac with local models, and 1.2 s or less with a hosted LLM. Barge-in stops TTS in under 300 ms.
Accuracy	Zero amounts or dates that disagree with bank data (checked in tests). Policy answers come from retrieved text only.
Privacy and security	The agent has no direct database access; the gateway is least-privilege and scoped per call. PII is masked in logs. All data is synthetic. Secrets live in a `.env` file, not in code.
Auditability	Every call can be fully reconstructed: audio, transcript, tool calls, sources and versions.

Capacity (POC)	One or two concurrent calls on a 16 GB M4 Mac.
Portability	Components are containerised or scriptable, so the move to a server or cloud and real SIP trunks needs configuration changes only.

10 Success metrics and acceptance criteria

- **Task success:** at least 80% of scripted scenarios reach the correct outcome (correct reason code, promise captured, or correct escalation).
- **Guardrails:** 100% of calls with account data had verification first; 100% of calls have the required disclosures; zero calls placed outside the window or over the cap in tests; 100% of disputes and cease-contact requests honoured.
- **Grounding:** at least 95% of policy answers judged faithful to the cited document; zero fabricated figures.
- **Experience:** latency targets met at P50; a human reviewer rates at least 4 out of 5 calls as "natural and respectful".
- **Demo:** live end-to-end inbound call, outbound missed-payment call, and warm transfer to a human softphone.

11 Milestones (indicative)

WEEK	DELIVERABLE
1	Mock bank (schema, seed data, API), policy knowledge base ingested, data gateway with tools and audit.
2	Local voice pipeline in browser test mode; inbound flows U3 and U4 working against the gateway and knowledge base.
3	SIP telephony (Asterisk and softphones); outbound campaigns U1 and U2 with the compliance gate; hand-off U6.
4	Console, tracing, simulated-caller test suite, tuning against the metrics, and the demo.

12 Risks and open questions

- **Local LLM quality:** an 8B-parameter model may be weaker at tool calling and following the flow. Mitigations: a strict state machine, few tools per state, and a hosted LLM switch.
- **Latency on a laptop:** Whisper is not a true streaming model. Mitigations: VAD-segmented transcription, smaller models and sentence-level TTS streaming.
- **Jurisdiction:** which regulator's rules (disclosures, calling hours, frequency caps) should the policy configuration model? US rules are used as the placeholder.
- **Voice and persona:** the bank name, agent name, tone, and whether the agent may negotiate amounts or only record them.